

1. INTRODUCTION

The space industry has progressed rapidly since inception due to demanding, diverse, and complex space missions. This evolution requires materials used by space systems to operate in the harsh space environment. Composite materials have emerged as strong candidates for space applications. Carbon fibre reinforced polymer (CFRP) is widely used in aerospace and cryogenic systems because of its high strength-to-weight ratio and favourable thermal properties[1].

Improved characterization of CFRP material properties at cryogenic temperatures, however, will enable greater and more advanced application of these materials to space environments. The objective of this work is to develop and validate methods to characterize material strength (i.e., Young's modulus), thermal expansion and conductivity, and heat capacity under cryogenic conditions. We present room temperature validation of some of this work and plans to extend this to cryogenic measurements.

2. EXPERIMENT TEST SETUP

A cantilever beam configuration was used to excite and measure flexural vibrations in material samples[2]. Samples were mechanically excited and their response was measured using a non-contact magnetic pickup sensor (i.e., Faraday Induction). The measured signal contains the beam vibration information required for extraction of the observed oscillation frequencies. The validated room-temperature methodology will be transferred to a Cryomech platform for measurements down to cryogenic temperatures (4 K).

Multiple samples can be measured within a single cooldown cycle to allow for control samples or multiple CFRP fiber orientations to be tested in parallel, enabling efficient material characterization. The top and bottom panels of Figure 1 show the room temperature and cryogenic test configurations.

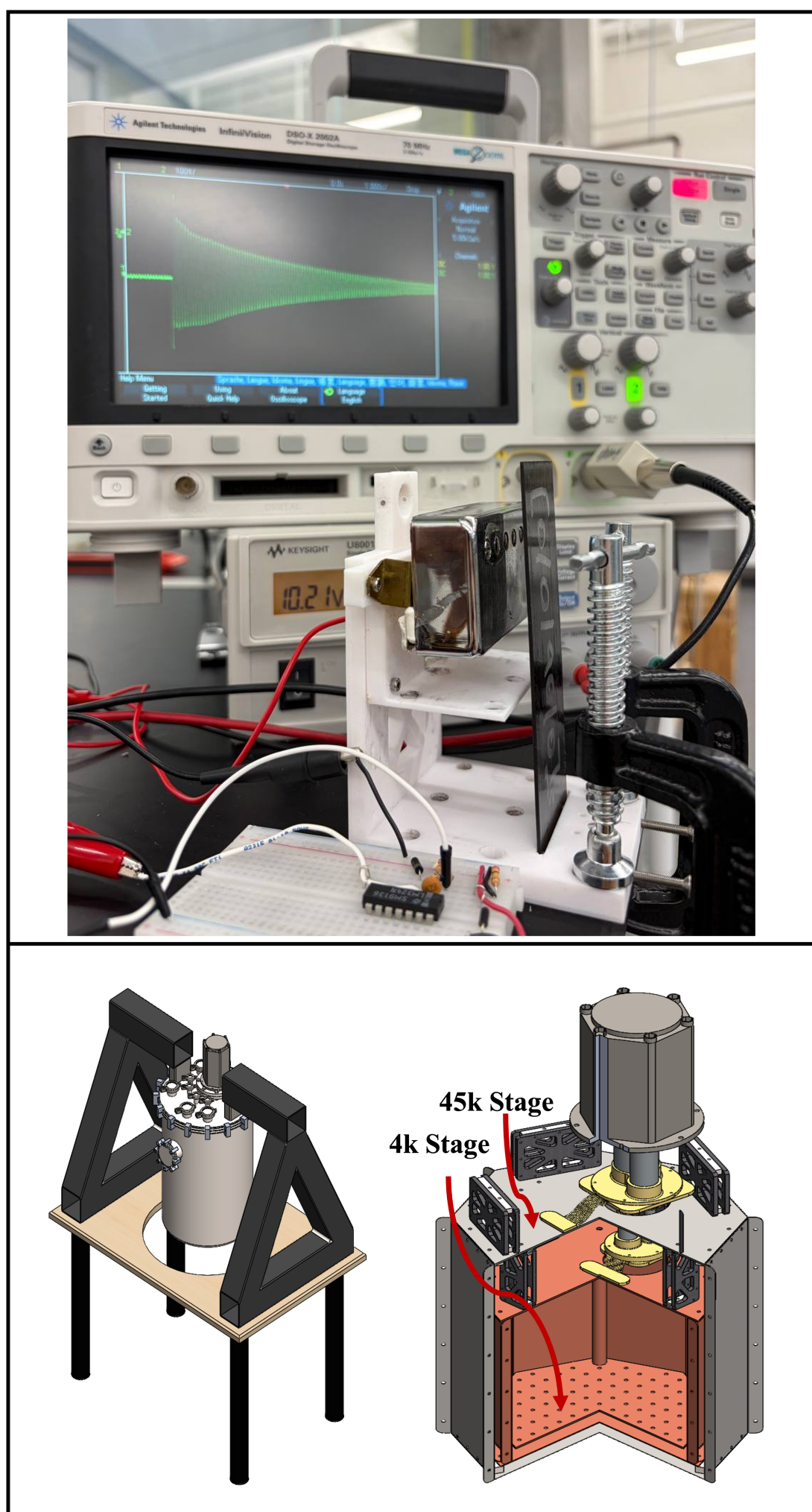


Figure 1: Top: Room-temperature setup. Bottom: Cryostat CAD design.

3. EXPERIMENTAL WORKFLOW AND SIGNAL PROCESSING

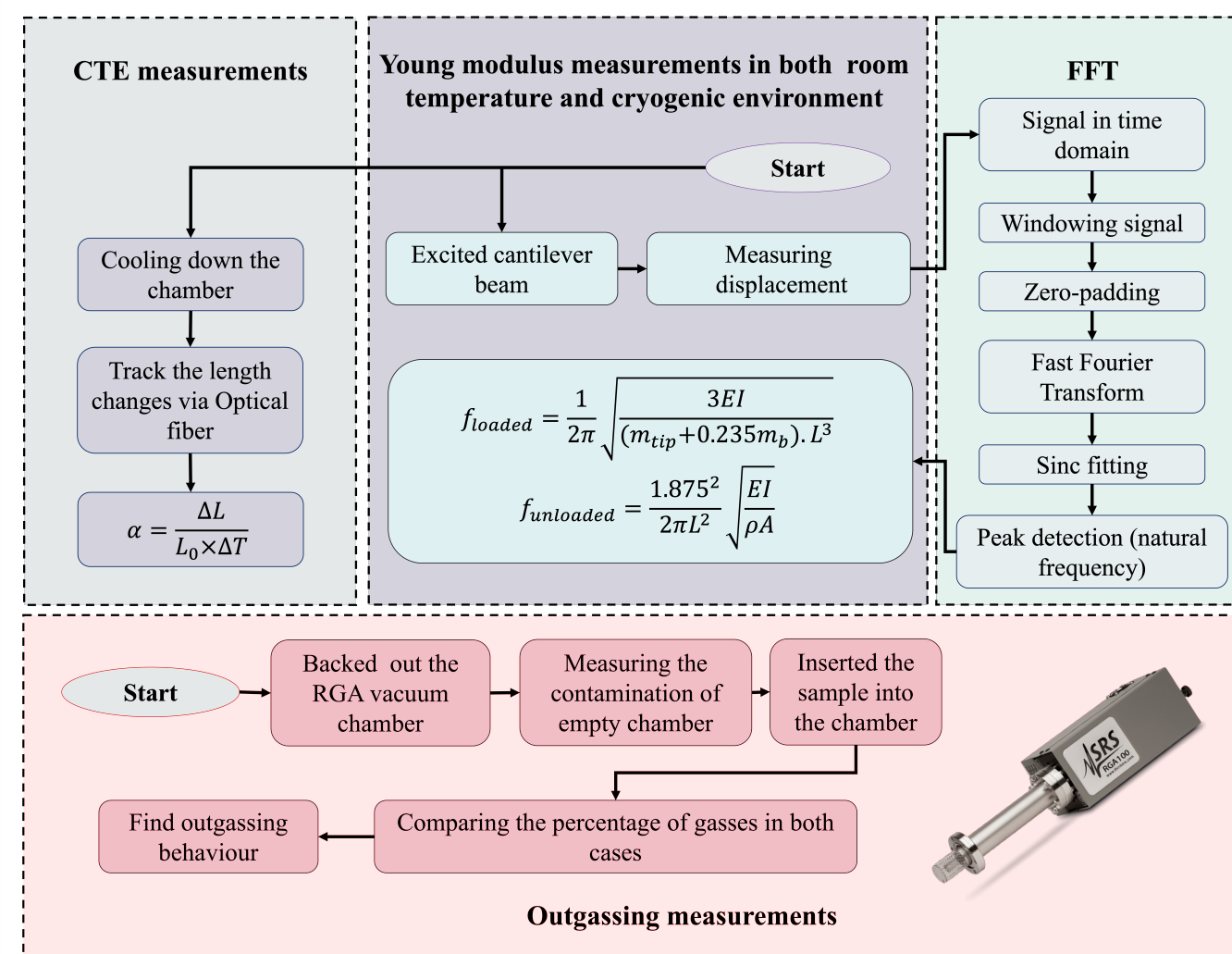


Figure 2: Project workflow.

In the shown flowhart, α represents material CTE, L_0 is the initial beam effective length, ΔL is the manipulated length change, ΔT is the temperature change, E is the Young modulus, I is the second moment of inertia for a given sample geometry, m

is the beam mass, L is the effective beam length, ρ is the material density, A is the beam cross section area, f_{loaded} and f_{unloaded} are the observed beam oscillation frequencies under loaded and massless conditions, respectively. Measurements were performed using both loaded and unloaded cantilever approaches. Frequency-domain analysis was applied to extract resonance frequencies, which were then converted into Young's modulus using beam vibration theory [3, 4]. Two independent methods were employed to estimate Young's modulus and provide internal validation of the measurement process. Time-domain signals were processed and evaluated in the Fourier domain. Monte Carlo uncertainty propagation techniques were employed to provide an estimate of uncertainty propagation.

The Young modulus and coefficient of thermal expansion (CTE) will be measured in the cryostat in parallel for multiple samples. We will also measure material properties as a function of the degree of cure by measuring vacuum outgassing between thermal cycles using an RGA 100 system [5].

4. RESULTS

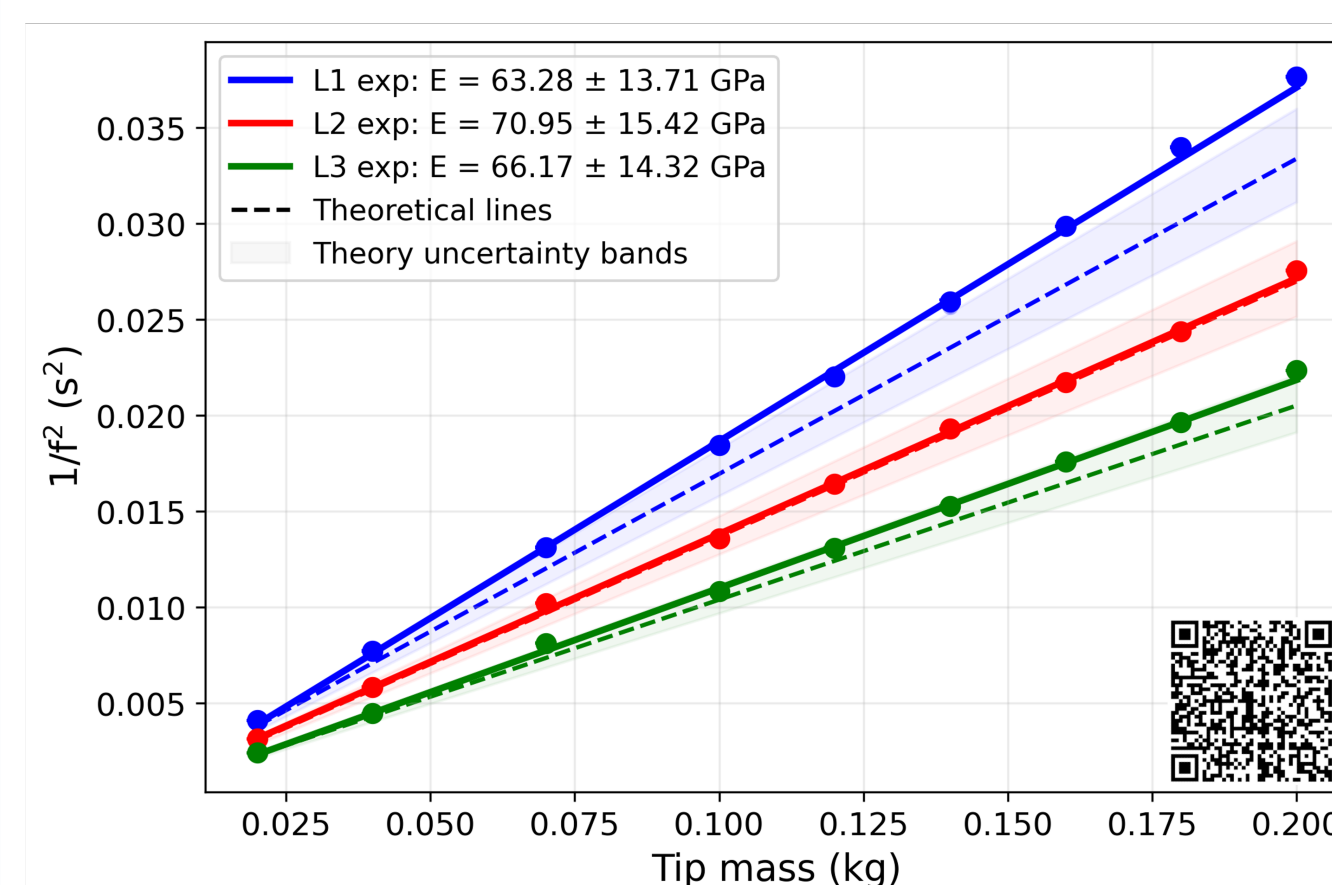


Figure 3: Aluminum results at room temperature: Loaded method. The results for other materials are shown in the QR code.

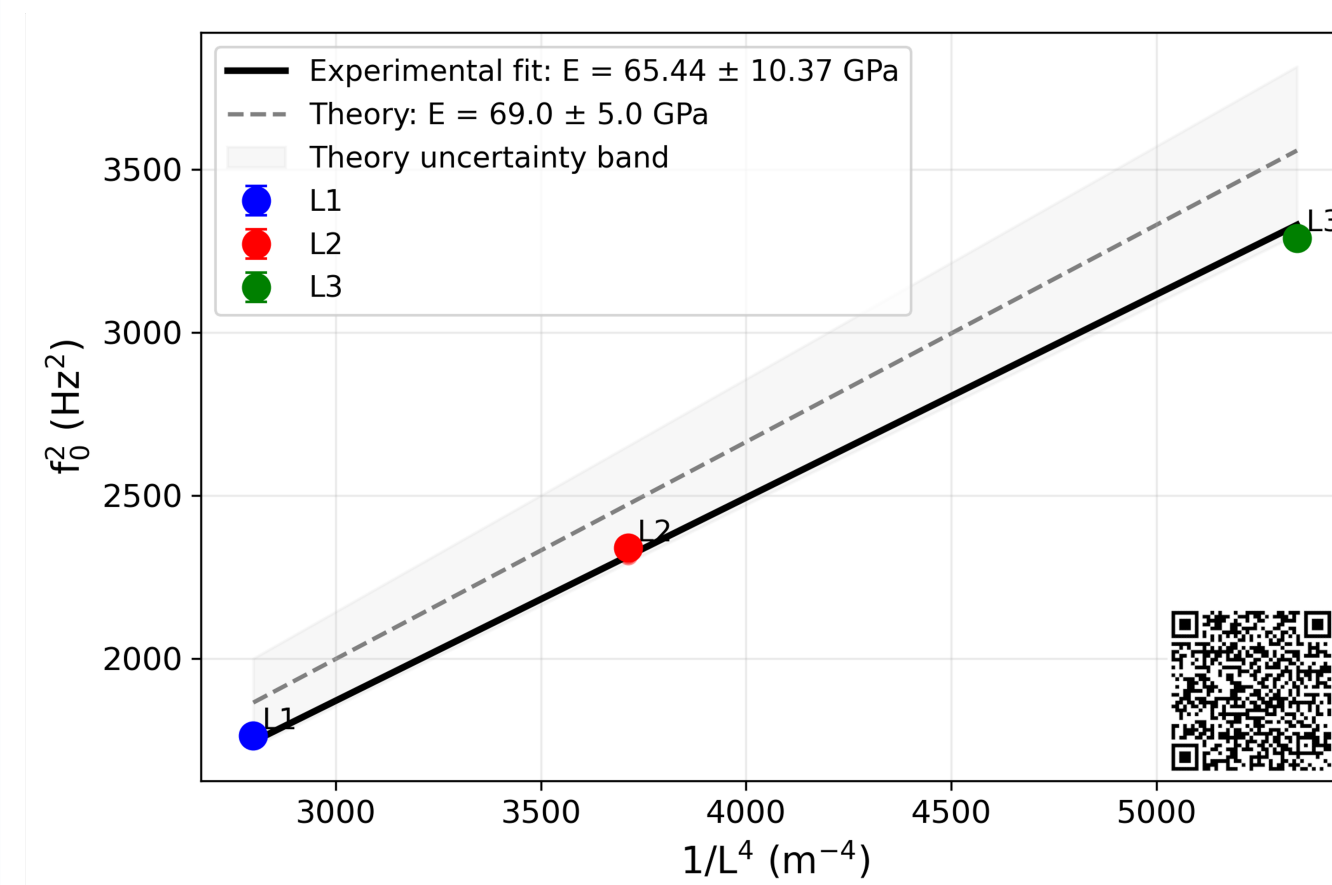


Figure 4: Aluminum results at room temperature: Unloaded method. The results for other materials are shown in the QR code.

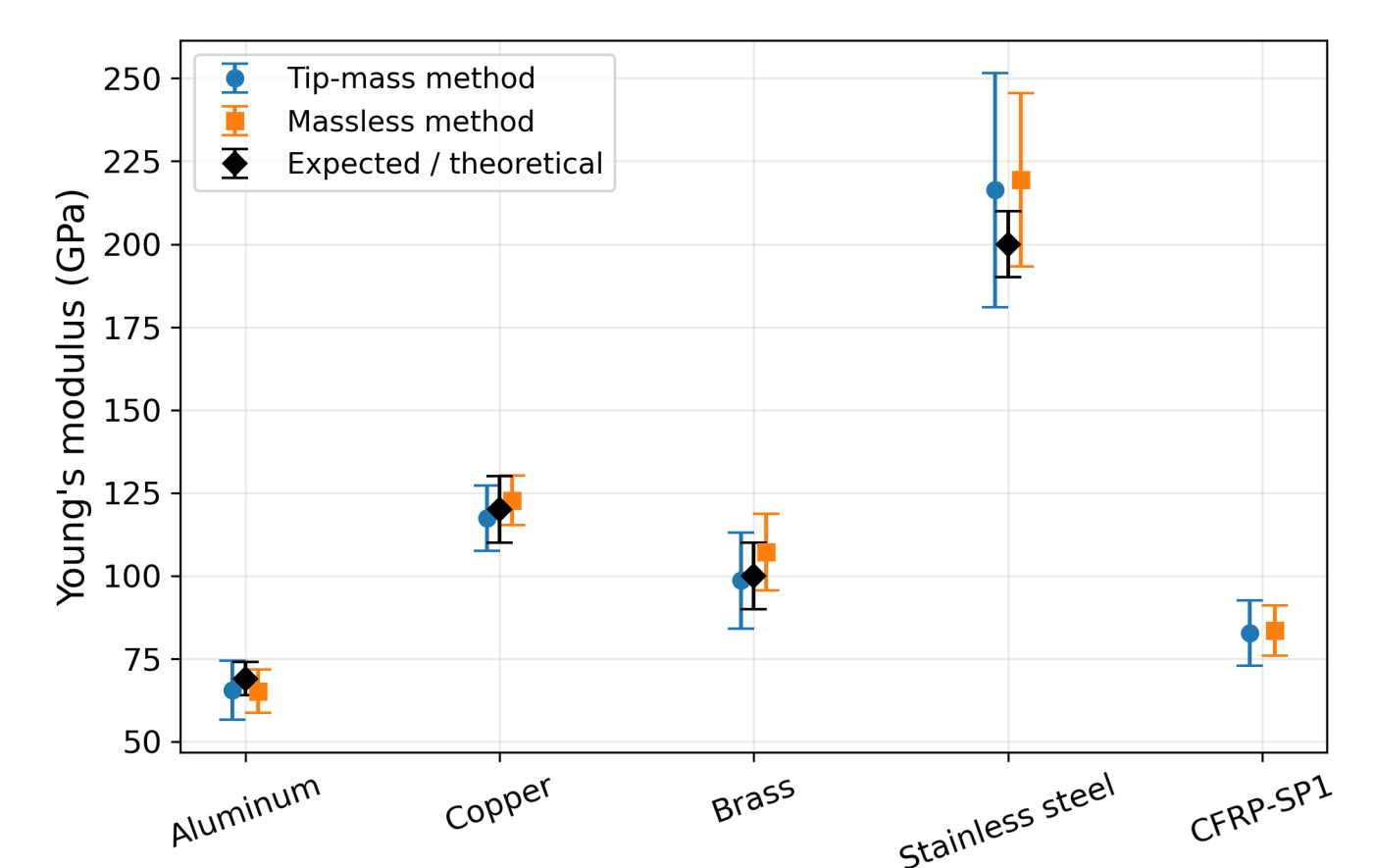


Figure 5: Comparison of Young's modulus across different materials obtained using tip-mass and massless methods. Theoretical values are shown for reference.

Figure 3 shows linear trends for different lengths for the tip-mass method. It can be seen that almost all cases all within uncertainty range where calculated from theoretical relations. The slope shows $\frac{4\pi^2 L^3}{3ET}$. Figure 4 illustrates a f_0^2 versus $1/L^4$ with a linear fit for massless beam. The slope is equal to $\frac{E}{\rho}$. Young's modulus measurements obtained from tip-mass and massless methods are shown in Figs. 3-5 with good between expected and measured values for known samples. Agreement between independent approaches validates the measurement and analysis pipeline. The validated room-temperature framework will be transferred to a similar cryogenic measurement. Uncertainties are more originated in the sample dimensions measurements.

5. CONCLUSIONS

A complete vibration-based framework for determining Young's modulus was successfully validated at room temperature. Our key findings are:

- Young's modulus successfully extracted for 5 materials
- FFT + sinc fitting improved frequency estimation
- Agreement observed between loaded and unloaded methods
- CFRP room-temperature baseline established
- Framework ready for cryogenic measurements

References

- [1] Locke D. Spencer et al. Evaluation of composite materials at cryogenic temperatures. In *Canadian International Conference on Composite Materials (CANCOM)*, 2019.
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- [5] Stanford Research Systems. *Models RGA100, RGA200, and RGA300 Residual Gas Analyzer: Operating Manual and Programming Reference*, 2009.